Multi-Horizon Industrial Electricity Procurement under Wholesale Price Dislocation

Adaptive Deep Reinforcement Learning Augmented with LLM-Derived Macroeconomic News Sentiment | MIBEL Backtest

1. Industrial Problem

Wholesale Market Exposure

- Iberian Market (MIBEL)
- Daily SPOT pool + 13 OMIP forward contracts (D, W, M)
- 24 MWh/day baseload demand

2026 Price Dislocation

−34.6% Mean Price Deflation
(67.6 → 44.2 €/MWh)

KS: D = 0.3653 ($p \leq 0.0002$)

Operational Setup

- 102 official OMIP trading sessions
- 46 non-trading dates (default SPOT)
- Physical delivery inventory vector

2. Multimodal AI Inputs

Physical & Derivatives State

- Grid: Solar, Wind, Demand, Hydro, Thermal, Nuclear
- Forward spreads: D1–D3, W1–W4, M1–M6 basis curves
- Strictly lagged observations

Zero-Shot LLM Sentiment

Local Meta Llama-3 8B (Ollama)
Strict JSON Schema Extraction
10 Fine-Grained Dimensions:
Short/Medium Term, Geopolitics, Renewables, Gas/CO2, Regulatory

Representation Ablation

- 2 × 2 Factorial validation (2025)
- Physical vs. Commodities vs. NLP
- M8 Hybrid isolates orthogonal signal

3. Sequential DRL Policy

Recurrent Deep RL Architecture

- RecurrentPPO (LSTM policy)
- Attention Transformer (GTrXL)
- 14 discrete purchase actions
- Explicit physical delivery tracking

Rolling Walk-Forward Protocol

Monthly Sliding Retraining
(W = 1 Month, 40 Test Blocks)

20-day hidden-state warm-up buffer
Evaluates strictly out-of-sample days

Market Friction Evaluation

- Linear fees: 0.10 to 0.25 €/MWh
- Friction penalty: 0.13 to 0.51 pp
- Net simulated savings remain positive

4. Savings–Risk Trade-Off

Simulated Cost Savings

- Multi-Year (374 days): +6.04% (Avg. 3.97 €/MWh cost reduction)
- Early 2026 Walk-Forward: +6.32% (±11.60% across 5 seeds; 44 days)

Downside Risk Mitigation vs. D2

Semi-Deviation: 13.68 → 9.15 €/MWh
(−33.1% downside volatility)

Max Drawdown: 225.5 → 186.5 €/MWh
(−17.3% maximum peak-to-trough)

Engineering Takeaway

- Fixed D2 gains nominal savings (+11.2%)
- M8-W1 hedges volatile down-tail risks
- Validates dynamic risk–return trade-off